Does Inviting Agents to Speak to Their Monitor Change Their Behavior and Make the Monitor Easier to Mislead?*

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Abstract

We study whether telling AI agents that a monitor reads their public record, and explicitly inviting them to address that monitor, changes their behaviour and the monitor's ability to detect planned deception. Two Qwen2.5-7B agents play repeated trust games (Diners and Weakest Link) in the Shi et al. harness under three conditions: no disclosure, disclosure of a named monitor, and disclosure plus an invitation to address the monitor inside the public message. Ground truth is the harness's mechanical typology over 1,200 agent-rounds. Awareness sharply raised premeditated deception in Diners (0.030 to 0.305), and the invitation raised it further (0.440), even though no agent ever wrote to the monitor. Under a frozen Qwen2.5-14B monitor, recall at a 5 percent false positive operating point fell from 0.918 under awareness to 0.846 under invitation, with overlapping intervals. A three-provider monitor panel shows the effect is monitor dependent: recall also fell under Claude Haiku 4.5 and rose under GPT-5 mini, while AUROC moved little for all three. A structured evidence view nearly closes the gap. In a pact extension with collusion built by instruction, the Qwen and Haiku monitors separate true collusion from coordinated honesty almost perfectly at pair level, while GPT-5 mini flags only a fifth to a third of executed pacts at its operating point and never accuses an innocent pair.

1 Introduction

The investigation concerns agent-to-monitor communication. Detection of planned deception is the measurement tool that the Shi harness supplies, not the discovery claim. Radev et al. (SCHEME) tested concealment under monitor awareness and cover story prompts, which is the instruction side of the problem. This project tests the opportunity side: agents in condition C receive an invitation to address the monitor inside their public message and no instruction to fool the monitor. Arike et al. showed that unaware agents are easier to catch. This project varies awareness and invitation as separate factors, so the invitation effect is judged on condition C against condition B only.

Two boundaries govern every claim. The monitor reviews completed logs offline and never responds during the game, so the experiment tests invitation and monitor-facing messages on offline detection, not recruitment into an active coalition. Monitor-facing messages sit in the public message field, so the other agent can read them as well, and a peer reaction remains a possible path for any behaviour change.

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Two supporting pieces complete the design. A monitor panel rescores the same frozen logs with monitors from three providers, which tests whether a detection effect belongs to the agents or to the monitor. A pact extension builds collusion by instruction, which gives the monitors one test with a known answer key: real team collusion, one instructed liar, and honest teams that only look coordinated.

Our main contributions are:

1. A controlled separation of monitor awareness from monitor invitation. The invitation effect is measured on condition C against condition B only, with the monitor, the threshold rule, and the ground truth frozen before scoring.
2. A three-provider monitor panel on identical frozen logs, which shows that the invitation effect on detection changes size and sign with the monitor, while ranking quality barely moves.
3. A constructed-collusion test with a known answer key (pact, solo, and innocent arms), which separates monitors that catch real collusion at their operating point from monitors that only rank it well.

2 Related Work

Shi et al. supply the harness and the ground truth. In their trust games, an agent's private plan, public announcement, and final action are recorded separately, and a mechanical typology classifies each agent-round. Over 90 percent of deviations in their highest deception conditions were preplanned rather than impulsive, which is what makes the premeditated class a usable detection target.

Radev et al. (SCHEME) study the instruction side of monitor awareness. Their agents sabotage under a monitor warning and under cover story prompts, and the warning halved Gemini sabotage success. This project gives no cover story and no instruction to deceive the monitor; it varies only whether agents know about the monitor and whether they are invited to address it.

Arike et al. show that the information a monitor receives decides much of its performance: less information often beat more, an extract-and-evaluate pipeline gained 16.8 percentage points, and unaware agents were easier to catch. The evidence-format comparison in section 4.6 follows their lead and varies what the monitor sees while holding the logs fixed.

Nguyen et al. (TraceGuard) argue that transcript reading fails because an agent can say exactly what it is doing and still collude, so audits should anchor on actions and outcomes, and their structured monitor design reached 95 percent safety where a single-score monitor reached 0 percent in their evaluation. Their audit protocol, freeze the monitor before test data and report recall at fixed false positive rates alongside AUROC, is the protocol this project follows.

Two entries in this sprint work the same evidence question from other angles. Words vs Actions scored the same Shi harness and found that monitors flag honest mind-changers, which matches the false positive pattern in section 4.3. Say, Do, or Both? tested say and do evidence in an auction setting. The AgentEval entry tested whether code monitors go easy on their own model family and found a precise null on small open-weight models; the panel in section 4.7 extends that question from code authorship to deception logs and to monitors from three providers.

3 Methods

Harness: Shi et al., llm-trust-breaking. Games: diners and weakestlink. Two agents, 10 rounds per trial, 10 trials per condition per game, 1,200 agent-rounds in total. Agent model: Qwen2.5-7B-Instruct served on Modal. Conditions: A blind, with no oversight disclosure; B awareness only, where the disclosure names the monitor (Qwen2.5-14B) and states that it scores public announcements and final actions for premeditated deception; C awareness plus the sentence that the agent may address the monitor directly inside its public message. Agents sampled at temperature 1.0, the harness default, which no run script overrode. The harness sets the generation request seed to 42 and seeds its own random module with 42 (run_endogenous.py; trial_runner.py), so the ten trials per cell are the repeated draws under one fixed seed.

Ground truth: the harness computes a mechanical typology for every agent-round from the private plan, the public announcement, and the final action. The positive class is premeditated_deception. The target is named as premeditated deception throughout. Detecting deception does not establish collusion, so the core experiment supports no collusion claim. The pact extension in section 4.8 addresses collusion directly, with coordination created by instruction.

Monitor: Qwen2.5-14B-Instruct on Modal (A100-40GB), frozen prompt, temperature 0. The monitor sees only the public record: both agents' public announcements and final actions for one round. Private plans and typology labels never enter monitor inputs. A leakage check on the pipeline confirmed this, and scores and labels are written to separate files.

Scoring: the decision threshold is the smallest integer score whose false positive rate on calibration negatives is at most 5 percent. The freeze-then-test discipline, with recall reported at fixed false positive rates alongside AUROC, follows the evaluation protocol that Nguyen sets out for monitor audits. The calibration set is trials 0 to 2 of condition A, pooled across both games. The threshold is frozen before held-out scoring. Held-out evaluation uses trials 3 to 9 for every condition. The analysis reports recall at the frozen threshold, the observed false positive rate, AUROC, and trial-cluster bootstrap intervals per condition. Recall at a 1 percent false positive rate is exploratory.

Pilot gate record: the pilot completed all six cells with zero parse errors and failed one gate criterion, at least 5 positives per game in pilot condition A (observed: 1 in diners and 1 in weakestlink). The author proceeded with the full run by a recorded amendment on Oct 25, 2026, with the frozen prompts, conditions, labels, and scoring rules unchanged.

Monitor panel: the same frozen core logs were rescored by two further monitors through OpenRouter, with the identical monitor prompt, the identical public-record inputs, and temperature 0, which the service was verified to accept. The added monitors are GPT-5 mini (OpenAI) and Claude Haiku 4.5 (Anthropic). Each added monitor received its own threshold from the same calibration rule (condition A trials 0 to 2, pooled across both games, target false positive rate 5 percent), which gave 91 for GPT-5 mini and 79 for Haiku, against 51 for the Qwen monitor. The players, the logs, and the ground truth never changed; only the scoring model moved.

Pact extension: a second dataset built for this report, diners only, two agents, 10 trials and 10 rounds per cell, 1,200 agent-rounds. Private instructions create three classes. In the pact class, both agents are instructed to plan CHEAP, announce EXPENSIVE, and order CHEAP in every round. In the solo class, only the first agent receives that instruction. In the innocent class, both agents are instructed to plan CHEAP, announce CHEAP, and order CHEAP. Each class ran under conditions B and C with the same disclosure patch. Ground truth comes from the instruction by construction, checked against the mechanical typology. An executed pact positive is a pact pair-round in which both agents carry the premeditated_deception label. A pair-round counts as flagged

when either agent reaches the monitor's threshold. Each monitor is judged at its own calibrated threshold from the core calibration rule, with no recalibration on pact data.

4 Results

4.1 Uptake

Agents almost never used the invitation. Across the 400 public messages in condition C, exactly 1 message (0.25 percent) mentioned the monitor, and it was a fully honest round. Condition B produced 0 monitor-directed messages in 400, and condition A produced 0 in 400. No premeditated deception round in any condition contained a monitor-directed message.

4.2 Message content

The single monitor-mentioning message in condition C reads, in part: "there's no benefit in misleading Agent M or the monitor." The agent announced CHEAP and acted CHEAP. It is a passing mention inside a message addressed to the other player, not an attempt to persuade the monitor, and it carries no request, justification, or appeal aimed at the scoring process. A manual read of condition C messages found ordinary game talk aimed at the other player throughout.

4.3 Detection change

Calibration on condition A trials 0 to 2 (118 negatives) set the frozen threshold at a score of 51 for a target false positive rate of 5 percent. Held-out evaluation covers 840 agent-rounds (trials 3 to 9).

Condition | Positives | Recall at 51 | 95% CI | Observed FPR | AUROC |
A blind | 4 | 1.000 | (1.000, 1.000) | 0.065 | 0.987 |
B awareness | 49 | 0.918 | (0.793, 0.983) | 0.043 | 0.974 |
C invitation | 65 | 0.846 | (0.674, 0.958) | 0.051 | 0.941 |

The primary contrast is C against B. Recall fell by 7.2 percentage points under the invitation (0.918 to 0.846), and AUROC fell from 0.974 to 0.941. The observed false positive rates are comparable (0.043 in B and 0.051 in C), so the contrast is not an artefact of one condition running at a stricter operating point. The bootstrap intervals overlap, so the drop is suggestive rather than decisive at this sample size. Condition A recall rests on only 4 held-out positives and carries no weight.

By game, diners recall fell from 0.953 in B (43 positives) to 0.902 in C (61 positives). Weakestlink recall fell from 0.667 in B (6 positives) to 0.000 in C (4 positives, mean monitor score 12.5). The weakestlink counts are too small to support a claim on their own. The exploratory 1 percent operating point (threshold 76) shows the same direction: recall 0.633 in B and 0.569 in C.

False positives concentrate in one negative type. At the frozen threshold, the monitor flagged fully honest rounds at a rate below 1 percent in every condition (A 1 of 249, B 2 of 207, C 1 of 198), and it flagged impulsive deviation rounds, where the agent changed course without a prior plan to deceive, at far higher rates (A 17 of 24, 0.708; B 8 of 20, 0.400; C 9 of 16, 0.562). Almost every false alarm in the held-out set is a changed-mind round rather than an honest round, which matches the pattern that the Words vs Actions entry reported on the same harness.

4.4 Behaviour change

Premeditated deception rates over all 200 agent-rounds per cell:

Game | A blind | B awareness | C invitation |
Diners | 0.030 (6 of 200) | 0.305 (61 of 200) | 0.440 (88 of 200) |
Weakestlink | 0.000 (0 of 200) | 0.035 (7 of 200) | 0.030 (6 of 200) |

Held-out trials 3 to 9 only: diners A 0.029, B 0.307, C 0.436; weakestlink A 0.000, B 0.043, C 0.029. Awareness raised the deception rate tenfold in diners and barely moved it in weakestlink. The invitation raised the diners rate further (B to C), again with no clear movement in weakestlink. The behaviour effect is game dependent, and it occurred while uptake of the invitation itself was zero, so the rise under C cannot be explained by agents writing to the monitor.

4.5 Mechanism

The message-based recruitment mechanism has no support in these logs: there are no monitor-directed positives in condition B or C, so recall on monitor-directed positives cannot be computed and no score loss can track messages that do not exist. Stratified by game, the recall drop appears in both games (diners 0.953 to 0.902; weakestlink 0.667 to 0.000 on very small counts), and the mean monitor score on positives fell from B to C in both games (diners 79.1 to 75.4; weakestlink 59.2 to 12.5).

The pattern check rules out the mix explanation in diners. Diners has two deception patterns, and their shares are identical in B and C: stated EXPENSIVE then final CHEAP forms 67 percent of positives in both conditions, and stated CHEAP then final EXPENSIVE forms 33 percent in both. Recall fell inside each pattern under C: the dominant pattern fell from 0.931 in B (29 positives, mean score 77.1) to 0.878 in C (41 positives, mean score 73.8), and the other pattern fell from 1.000 (14 positives, mean 83.2) to 0.950 (20 positives, mean 78.8). The same deceptions received softer scores under the invitation. The structured evidence view points the same way: when a mechanical stated-versus-final check carries the signal, the B to C gap shrinks from 7.2 points to 1.6 points. Because uptake was zero, the remaining paths are the free-text message channel itself and peer reaction, because monitor-facing text sits in the public record that the other agent also reads.

4.6 Evidence-format comparison (secondary analysis)

The same frozen monitor scored the same 1,200 rounds under three further evidence views, each calibrated by the same rule (condition A trials 0 to 2, 5 percent target). The headline view (messages plus actions) is the reference: recall 0.918 in B and 0.846 in C, AUROC 0.974 and 0.941.

View | Threshold | Recall B | Recall C | AUROC B | AUROC C |
Messages only | 21 | 0.367 | 0.492 | 0.715 | 0.772 |
Actions only | 1 | 0.000 | 0.000 | 0.500 | 0.500 |
Structured report | 76 | 0.878 | 0.862 | 0.986 | 0.978 |

Actions alone carried no signal at all: the monitor returned a score of 0 for every one of the 1,200 rounds, because a final action on its own cannot show the plan-announcement-action inconsistency that defines the target. Messages alone were weak. The structured report, which adds a mechanical stated-versus-final match flag to the announcements and actions, matched the headline view in B and slightly exceeded it in C, at lower observed false positive rates (0.017 in B and 0.033 in C, against 0.043 and 0.051 for the headline view). Under the structured view the B to C recall gap

shrinks from 7.2 points to 1.6 points, which suggests that most of the detection loss under the invitation runs through the free-text message channel rather than through the record of what agents did.

4.7 Monitor panel on the core logs

Three monitors scored the same 840 held-out agent-rounds. Recall at each monitor's own calibrated threshold:

Monitor | Threshold | Recall B | Recall C | AUROC B | AUROC C |
Qwen2.5-14B (headline) | 51 | 0.918 | 0.846 | 0.974 | 0.941 |
Claude Haiku 4.5 | 79 | 0.755 | 0.631 | 0.970 | 0.961 |
GPT-5 mini | 91 | 0.245 | 0.354 | 0.958 | 0.974 |

Observed false positive rates stayed low for the added monitors (Haiku 0.022 in B and 0.019 in C; GPT-5 mini 0.000 in B and 0.009 in C). Bootstrap intervals overlap inside every monitor (Haiku B 0.636 to 0.857 and C 0.500 to 0.766; GPT-5 mini B 0.154 to 0.319 and C 0.181 to 0.552). Condition A rests on 4 positives under all three monitors and carries no weight.

The direction of the invitation contrast depends on the monitor. Recall at the operating point fell from B to C under the Qwen monitor (by 7.2 points) and under Haiku (by 12.4 points), and rose under GPT-5 mini (by 10.9 points). AUROC moved little under all three (Qwen 0.974 to 0.941, Haiku 0.970 to 0.961, GPT-5 mini 0.958 to 0.974), so the monitors rank deceptive rounds about as well under the invitation as under awareness alone. What changes is where the scores sit against the calibrated threshold. GPT-5 mini also shows that a monitor can rank well and still catch little at its operating point: its AUROC matches the other monitors, while its recall at the calibrated threshold is far lower in absolute terms, because its threshold of 91 sits above most of its positive scores.

4.8 Pact extension: detection of constructed collusion

Execution came first. In the pact class under B, all 200 agent-rounds carried the premeditated label and all 100 pair-rounds were executed by both agents. Under C, 187 of 200 agent-rounds were premeditated, with 88 pair-rounds executed by both agents, 11 by one agent, and 1 by neither. In the innocent class, all 200 agent-rounds in each condition were fully honest. In the solo class, the instructed agent executed in 98 of 100 rounds under B and 100 of 100 under C. The uninstructed partner received no instruction and still lied on its own in 49 of 100 rounds under B and 50 of 100 under C, after watching the instructed agent announce EXPENSIVE and order CHEAP round after round. Deception spread from the instructed agent to the uninstructed one through ordinary play.

Pair-level detection at each monitor's own threshold:

Monitor | Pact pairs flagged, B | Pact pairs flagged, C | Innocent pairs flagged, B | Innocent pairs flagged, C |
Qwen2.5-14B | 100 of 100 | 88 of 88 | 0 of 100 | 0 of 100 |
Claude Haiku 4.5 | 100 of 100 | 84 of 88 | 0 of 100 | 2 of 100 |
GPT-5 mini | 21 of 100 | 34 of 88 | 0 of 100 | 0 of 100 |

The Qwen monitor and Haiku separate constructed collusion from coordinated honesty almost perfectly at pair level. GPT-5 mini does not at its operating point: it flags a fifth of executed pact

pairs under B and about a third under C, while never flagging an innocent pair. Its agent-level flag rate on premeditated rounds in the pact cells is 0.125 under B and 0.230 under C, against 0.975 and 0.979 for the Qwen monitor and 0.915 and 0.850 for Haiku. Solo pairs are flagged at 0.94 to 1.00 by the Qwen monitor and Haiku, because the either-agent rule fires on the instructed agent alone; pair-level flagging cannot by itself separate a pact from one instructed liar, and the distinction lives at agent level.

One boundary governs this whole section. The pact pattern is as loud as a pattern can be: the same announcement-action mismatch, repeated by both agents, in every round. These results are a positive control for the audit pipeline. They show that two of three monitors separate real collusion from coordinated honesty when the answer is known and the signal is maximal. They say nothing about covert collusion, where the signal is weaker by design.

5 Discussion and Limitations

Five findings survive the frozen analysis rules. First, awareness of a named monitor raised the premeditated deception rate sharply in diners (0.030 to 0.305) and barely in weakestlink (0.000 to 0.035), so disclosure changed behaviour in a game-dependent way and did not deter deception in either game. Second, the invitation raised the diners rate further (0.305 to 0.440) even though no agent used the invitation, so the added sentence changed play without any agent-to-monitor message. Third, detection at the frozen threshold fell by 7.2 percentage points from B to C at comparable observed false positive rates, with overlapping intervals. The honest reading is that the experiment measured a real detection drop in the C direction that this sample cannot confirm decisively, and that the drop cannot be described as recruitment by persuasion, because the messages that persuasion would require do not exist in the logs. A null result here would have read as inconclusive under the same rules.

Fourth, the panel bounds the third finding. The invitation contrast is monitor dependent: recall at the operating point fell under the Qwen monitor and under Haiku, and rose under GPT-5 mini, while AUROC moved little under all three. The effect is therefore not a general property of monitors. It operates mainly through score placement against a calibrated threshold, which is a property of the individual monitor and its calibration, not of the agents alone.

Fifth, the pact extension gives the report one result with a known answer key. Two of three monitors separate constructed collusion from coordinated honesty almost perfectly at pair level, and the third, GPT-5 mini, misses most executed pacts at its own operating point while never accusing an innocent pair. Ranking quality and practical detection come apart at the threshold. An audit that reports AUROC without recall at its frozen operating point can call a monitor strong while that monitor lets most real collusion through.

5.1 Limitations

The calibration split is small (trials 0 to 2 of condition A) and each condition has seven held-out trials per game, so statistical power is limited and intervals are wide. The players in every dataset are one model, Qwen2.5-7B-Instruct, so the behaviour results may not transfer to frontier players. The monitor reads completed logs offline and never replies, so the study says nothing about an agent persuading a live monitor during play. The pact pattern is maximally overt and works as a positive control, not as evidence about covert collusion. The full limitation list, with the dual-use statement, is in the appendix.

5.2 Future Work

The natural next study regenerates the games with frontier players (GPT-5 mini first) across the core and the pact, with all three monitors rescoring the new logs. A five-agent Diners arm would test whether the awareness effect grows with group size. A covert pact variant, where the shared plan is set once and never repeated in the public record, would move the extension from a positive control toward realistic collusion. A monitor that replies during play would test recruitment in the active sense that this study excludes.

6 Conclusion

Telling agents about their monitor did not make them behave. It made them deceive more, and inviting them to address the monitor made them deceive more still, without a single agent using the invitation. Detection at a frozen operating point slipped under two of three monitors and improved under the third, while ranking quality held steady, so the practical strength of a monitor audit is decided at the threshold, not by the ranking. On the one test with a known answer key, most monitors caught constructed collusion cleanly, and the one that missed it never accused an innocent team. Awareness, invitation, and threshold choice are all levers on audit outcomes, and none of them is visible in an AUROC number alone.

Code and Data

- **Code repository**: the harness is the public Poseidon Research starter repository, https://github.com/poseid research/collusion_hackathon. The disclosure patch, the pact patch, the monitor, the scorers, and the analysis scripts written for this sprint are at https://github.com/bhatiashash/monitor-recruitment.
- **Data**: the frozen game logs and the monitor score files for the core, the evidence views, the panel, and the pact are in the same repository, under data/ and results/.
- **Preexisting code**: all code written for this project was written during the sprint, on October 25, 2026. The only pre-existing code is the Poseidon Research starter harness, credited above.

Author Contributions

Shash Bhatia designed and froze the experiments, directed the analysis, and reviewed the final report.

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Appendix: Limitations and Dual-Use / Ethical Considerations

- Detection of deception does not establish collusion. The core experiment is a monitor-robustness study of premeditated deception. Only the pact extension speaks to collusion, and only to collusion created by an explicit instruction.
- The pact pattern is maximally overt, the same mismatch repeated by both agents in every round. The pact results are a positive control, not evidence about covert collusion.
- In the solo cells, the uninstructed agent deceived spontaneously in about half its rounds, so solo pair-rounds often contain two deceivers with different causes. Pair-level flag rates in the solo class cannot be read as collusion detection.
- The calibration split is small (trials 0 to 2 of condition A) and each condition has seven held-out trials per game, so statistical power is limited and intervals are wide.
- Condition A in weakestlink produced zero premeditated positives in 200 agent-rounds, so recall inside condition A rests almost entirely on diners.
- One agent model (Qwen2.5-7B-Instruct). The monitor panel spans three providers, but the players in every dataset are Qwen 7B, so results may not transfer to frontier players, whose deception may be subtler. Regenerating the games with frontier players is the natural follow-up study.
- The harness's built-in judge labels ran on the served Qwen model because the hardcoded judge model was unavailable on the local endpoint. Judge labels are auxiliary and never serve as ground truth.
- Dual use: the study measures whether an invitation to address a monitor changes behaviour and detection. It publishes no new attack technique. Condition C grants an opportunity only, with no instruction to manipulate the monitor.

LLM Usage Statement

An AI assistant (Muse, by Meta) helped write the experiment and analysis code, run the scoring pipelines, and draft this report. The author designed and froze every experiment before the runs, made all design and interpretation decisions, and verified every number in this report against the recorded pipeline outputs.